

Xiaotong Shen

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Education

University of Toronto

May 2027

Honours Bachelor of Science (HBSoc) in Data Science, Computer Science + ASIP Co-op

Major GPA: 4.0/4.0, cGPA 3.75/4.0

Relevant Coursework: Linear Algebra, Multivariable Calculus with Proofs, Software Design (Java), Probability and Statistics I & II, Computer Organization, Machine Learning, Data Structures and Analysis (C++), Data Science

Experience

Research Assistant, Software Developer

Toronto

COBWEB

August 2025 - Present

- Migrated existing simulation functionality from the legacy **Java** COBWEB application to a **Javascript/Typescript web-app** to enable browser-based execution and to improve accessibility to **15+ high-school researchers**.
- Implemented a **real-time data visualization software** with **WebGPU**, demonstrating how agents learn rule-based behaviors under a **Learning Classifier System** to enhance interpretability of agent-environment adaptations.
- Researched **advanced statistical and machine learning techniques** to validate simulation accuracy, drawing from ecological modeling and **process-evaluation frameworks** to ensure methodological rigor.

Developer & Researcher

Toronto

AURA - UTMIST (University of Toronto Machine Intelligence Student Team)

August 2025 - Present

- Developed **multimodal emotion/personality recognition pipelines** combining speech prosody and personality estimation models to increase accessibility in mental health treatment through conversational agents.
- Engineered and tested **response generation subsystems** with controllable language style, prosody planning, and coherence/safety mechanisms, to balance empathetic expression with responsible LLM practices.
- Collaborated on **full-system integration** and evaluation using metrics such as **System Usability Scale (SUS)** and **Conversational Turns per Session (CPS)**, iteratively improving model performance and user engagement.

Arrive Ready Intern – Data Engagement

Toronto

University of Toronto – Arrive Ready Internship

May 2025 – August 2025

- Collected and analyzed engagement data from **5,000+** student interactions (email open rates, event attendance) with **Python** to identify behavioral trends and recommend improved outreach strategies.
- Authored and presented a comprehensive end-of-term report on content engagement, with **5+ actionable recommendations**, informing future program design, data collection practices, and outreach strategies.
- Designed and published social media content for **@sidneysmithcommons** on Instagram, generating **100,000+ cumulative likes** to increase student awareness of programming.

Student Instructor

Remote

Stanford University (Code in Place)

May 2025 – July 2025

- Instructed weekly tutorials for a cohort of **20+** students on core **Python** concepts such as **data structures, functions, and memory management** as a part of **Stanford University's Code in Place** program.

Projects

Web App for Predicting TTC Delays | Javascript, Python

[Github](#) (2025)

- Built an **interactive data dashboard** using **Streamlit** and **Plotly** to visualize real time TTC delay predictions, designing charts and reports that surface model outputs in a clear and accessible way.
- Performed data preprocessing and feature engineering on CSV transit datasets using **pandas**, trained and tuned a Random Forest classifier, and connected the model to the Streamlit front end for live predictions.

U of T Lost and Found Platform | Java, Firebase API

[Github](#) (2025)

- Designed and implemented a scalable database driven **Java** application using **Object Oriented Programming** and SOLID principles, consuming **Firebase REST APIs** for real time data storage and retrieval across multiple user flows.

Technical Skills

- **Programming Languages:** TypeScript, Javascript, Python, Java, R, SQL, HTML/CSS, C++
- **Frameworks & Tools:** React, Node.js, TensorFlow, PyTorch, Git/GitHub, Docker, AWS, Unix/Shell Scripting
- **Databases & APIs:** MySQL, PostgreSQL, MongoDB, SQLite, REST APIs, GraphQL
- **Software Development:** Object-Oriented Programming, Data Structures & Algorithms, Agile, CI/CD